CAD Readers and Data Exchange

As of JobLine G-Code for VS Code 0.3.12

Models and setup data

CAD System / Data	Format	Version	Read	Write	Feature / Data Exchange	Sketch / 2D Data
STEP	*.stp, *.step	AP203 / AP214 / AP242 (ISO 10303-21)	✓[1]			
STL	*.stl	ASCII and binary	✓			
Parasolid (text)	*.x_t, *.xmt_txt, *.xmt	Text transmit files	✓[2]			
Parasolid (binary)	*.x_b	—	[3]			
Autodesk Fusion setup	*.f3d	Fusion archive (ZIP)	✓[4]		✓[4]	
JobLine machine	*.jblmachine (JSON)	Format 1.0	✓	✓	✓[5]	
RevPack	revpack.json	`jobline.revpack/2`	[6]		[6]	
RevPack (v1 manifest)	manifest JSON	v1 (gcode + goal models)	[7]			
Lathe insert codes	ISO designation	Shapes C, D, R, S, T, V, W	✓			

[1] Tessellated through OpenCascade (occt-import-js 0.0.23, at most 300,000 triangles) and shown as a scene or target model. The schema name comes from FILE_SCHEMA. Features, PMI and assembly structure are not read.

[2] Bounds only: every 3D point in the file is collected and the bounding envelope is drawn. Surfaces are not rebuilt.

[3] The import dialog lists *.x_b, but the file is read as text, so binary Parasolid is not decoded. Treat it as not supported and export *.x_t or STEP.

[4] Reads the active setup's stock size and allowances, work offset, WCS orientation and origin, operation names, the model envelope and the preview PNG. The solid model and toolpaths are not read.

[5] `.jblmachine` is byte-identical in the extension, jobline.ai-CAM and the JobLine.ai platform (`src/presets/jblMachine.ts`), so a machine exported from one opens in the others.

[6] In draft PR #7, not released (0.3.14 on that branch): RevPack v2 loads target, stock, workholding, tools and machine. STEP targets go through OpenCascade and are cross-checked against the paired STL. Its AR4 real-part checks compare STEP and STL digests from local, gitignored Annin AR4 files.

[7] Development harness only (`npm run visualizer:harness:revpack`), not a user command.

CAD Readers and Data Exchange (continued)

G-code and control dialects

JobLine's equivalent of a CAD reader: the program formats the extension opens, checks and explains. Any G-code file also gets highlighting, the toolpath visualizer, the explainer and the Toolbox edits.

Control	Program files	Models named in the app	Read	Validate	Post	Dialect-specific parsing
Fanuc	*.nc, *.tap, *.sub, *.lib	0i / 30i / 31i	✓	✓[8]	[9]	Macro B variables and expressions. In draft PR #9: Fanuc G list, variable ranges, WHILE/DO/END and GOTO targets, O-number and M-per-block limits
Haas	*.nc, *.5ax, *.ncc, *.ncp, *.ncf, *.nc1	NGC	✓	✓[8]	[9]	Fanuc-style; Haas Classic not separate. In draft PR #9: Haas G list (G47, G150, G187, G234…), M97 P→N targets, 5-digit O
Siemens Sinumerik	*.mpf, *.spf	840D / 828D	✓	✓[8]	[9]	CYCLExx, MCALL, HOLES1/2, POCKET1–4, SLOT1/2, LONGHOLE, SWIVEL; R-parameters. In draft PR #9: DEF placement, IF/WHILE/FOR/LOOP/REPEAT pairing, GOTOF/GOTOB labels, R range, MCALL left on, ISO-isms, G291/G290
Okuma OSP	*.min, *.prg	OSP-P	✓	✓[8]	[9]	$ program headers, #V variables, CALL OBn; lathe detection. In draft PR #9: V-not-# variables, G56 tool length, N-label jumps, RTS on called subs
Mazak	*.eia, *.iso	Smooth / Matrix (EIA/ISO)	✓	✓[8]	[9]	EIA/ISO G-code only. In draft PR #9: Mazak G list, 8-digit O; Mazatrol detected and skipped, not read
Fadal	*.fgc	Format 2	✓	✓[8]	[9]	In draft PR #9: Fadal macros (#V1=…, #IF … THEN GOTO :LABEL), V1–V100, label targets
Generic / LinuxCNC	*.gcode, *.ngc, *.g, *.cnc, *.dnc	—	✓	✓[8]	[9]	In draft PR #9: O-word sub/if/while/repeat structure and call targets
Heidenhain TNC	*.h (Klartext), *.i (ISO; add .i under JobLine additional extensions), *.cyc, *.tab	Klartext / ISO	✓[10]	✓[10]	[9]	In draft PR #9: Klartext BEGIN/END PGM, block numbers, TOOL CALL axis, L/C/CC/CR, Q0–Q1999, LBL / CALL LBL; ISO %name G71 header and footer. Released 0.3.12: family name only
Mitsubishi	*.nc	M8 / M80	✓	✓[10]	[9]	In draft PR #9: Mitsubishi G list (G66.1, G120.1…), variable ranges, up to 4 M per block. Released 0.3.12: family name only
Fanuc Robot	*.ls, *.tp	TP / LS	✓[11]			Language and control entry; no robot kinematics
ABB RAPID	*.mod, *.pgf	RAPID	✓[11]			Language and control entry; no robot kinematics
KUKA KRL / CL data	*.src, *.ptp, *.apt, *.cls	—	[12]			Opened as G-code; KRL and CL/APT are not interpreted

[8] Released 0.3.12: one generic ISO/Fanuc-style rule set for every control (canned-cycle Z/R/Q, arc validity, spindle and coolant at program end, axis travel against the machine preset). In draft PR #9: each control also gets its own rule set, listed per row; unknown G-codes are information only because options differ by machine. Rules check syntax and structure, not execution, and are tested on JobLine-written fixtures, not certified on each control.

[9] No CAM post-processor; posting is in jobline.ai-CAM (Haas NGC mill). In draft PR #9: **Convert Program to Another Control** rewrites common codes from ISO dialects to Haas, Mazak, Mitsubishi, Siemens 840D (G81/G82 → CYCLE81/82, G43 → D), Okuma OSP (G54 → G15 H, G43 → G56) and Heidenhain Klartext (L/C/CR, G81–G83 → CYCL DEF 200). Macros, tapping/boring cycles and anything else it can't map are left as `JL MANUAL` comments. Converting from Siemens, Okuma or Heidenhain is not supported.

[10] In draft PR #9, Heidenhain and Mitsubishi are offered in **Select CNC / Robot Control Type** and validated. Klartext programs skip the ISO rules. *.h is already treated as G-code by file name; `0 BEGIN PGM` is recognised as a G-code first line.

[11] Syntax highlighting (shared G-code grammar) and a control entry for preset matching only.

[12] Registered as G-code extensions, so they get highlighting and the generic G-code reader only.

CAD Readers and Data Exchange (continued)

Tool libraries and CAM output (in draft PR #9)

Not released. Built on branch feat/unbuilt-pieces-2026-10-15 (draft PR #9). Imported tools feed the Tools view and the checks: tools not in the library, T/H and T/D register mismatches, comment diameter vs library diameter.

Source	Format	Fields read	Read	Write	Linked to program	Status
Fusion tool library	*.json, *.tools	T, H, D, DC, LCF, gauge length / LB, OAL, RE, unit	✓[13]		✓	In draft PR #9
HSMWorks / Inventor CAM	*.hsmlib (UTF-16 XML)	nc number, length / diameter offset, body diameter, flute, body and overall length, corner radius	✓[13]		✓	In draft PR #9
Generic tool list	*.csv, *.tsv	T, H, D, diameter, flute length, gauge length, description (common spellings, Fusion CSV headers)	✓[13]		✓	In draft PR #9
CAM setup sheets	*.html, *.htm, *.xml, *.csv	Tool tables from ESPRIT, Mastercam, Fusion, GibbsCAM and similar sheets (column or per-tool layout)	✓[14]		✓[14]	In draft PR #9
Posted NC header	Program comments	CAM system name; tool list comments (Mastercam, Fusion style)	✓[14]		✓	In draft PR #9
Live CAM session / native CAM files	*.esp, *.mcam, *.vnc, .f3d toolpaths	—				Not built[15]

[13] Holders and probes are skipped. Tools with no number, and duplicate numbers (vendor catalogues often leave every tool at 1), are listed instead of imported.

[14] A sheet with the program's name in the same folder (1001.html, 1001_setup.xml, Setup Sheet 1001.htm) links automatically; otherwise the post's header tool list is used. Fixtures are written in the usual template shape, not real vendor exports; setup-sheet templates are user-editable, so real sheets may need new column names.

[15] A live link or native file reading needs each vendor's licensed SDK (ESPRIT API, Mastercam NET-Hook, GibbsCAM SDK, Fusion API inside Fusion, NX Open). Not possible from a VS Code extension without them.

⚠ **Notice: Not in the 0.3.12 release: tool-library import, CAM output integrations, per-control validation and dialect conversion are in draft PR #9 (not merged). No live CAM connection is planned without vendor SDKs. DXF, IGES, SolidWorks and other native CAD formats are not read.**